

Article

Horizontal Active Noise Control Based on Wave Field Reproduction Using a Single Circular Array in 3D Space

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Abstract: In this paper, we propose horizontal active noise control (ANC) using two-dimensional wave field information alone. By reducing the control space to a horizontal plane, the number of microphones and speakers was considerably reduced compared with ANC systems using three-dimensional wave field information. The radii of the reference, microphone, and loudspeaker array were determined based on the wave field reproduction error. Accordingly, the simulation and experimental results of the proposed ANC system were presented based on the use of five microphones and loudspeakers using conventional ANC algorithms. Overall, an average noise reduction of 20 dB was observed inside the microphone array with a radius of 0.5 m for tonal noise at 200 Hz. This performance is acceptable with a drastically reduced number of microphones and speakers. The findings of this study, along with further research conducted in a reverberant room, represent a significant contribution to global ANC commercialization.

Keywords: time-domain signal processing; active noise control; wave field reproduction



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1. Introduction

Low-frequency noise, which is challenging to reduce using passive methods, can be reduced using a transducer-like loudspeaker or devices that generate vibration. Active structural acoustic control (ASAC), which reduces the sound power radiated into the air by controlling the vibration of the noise source [1,2], and active noise control (ANC), which cancels the sound pressure by generating an “anti” signal that has an opposite phase but equal amplitude to those generated by the noise source, have been researched [3]. Given that ASAC reduces noise emission from a noise source, global noise reduction is possible; however, if the noise is radiated from several unspecified sources, an effective reduction is not possible unless the noise source is specified. However, because ANC eliminates sound pressure propagating in the air at the microphone position, it can be effectively used in situations where the noise source is unspecified [4,5]. Local ANC is achieved by reducing the sound pressure at the microphone position; however, the wave field of the noise source must be suppressed to achieve global ANC. To achieve this, acoustic energy is selected as the cost function, and an additional microphone has been used to obtain the particle velocity. As a cost function, acoustic energy is more efficient than acoustic pressure for global noise reduction [6]. In an intuitive manner, if an “anti” wave noise-source field is generated, global ANC can be achieved. Wave field reproduction provides an immersive experience to all listeners within a large target area by creating a virtual wave field of the sound source using a densely placed loudspeaker configuration. Wave field reproduction can be divided into wave field synthesis, based on the Kirchhoff–Helmholtz integral, and higher order ambisonics, which uses harmonics coefficients [7]. However,